



HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.



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FUNDAMENTALS OF OPTIMIZATION

Integer Linear Programming

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CONTENT

- Relaxation and Bound
- Branch and Bound
- Cutting plane
- Integer Rounding
- Gomory Cut
- Branch and Cut algorithm

RELAXATION AND BOUND

- Given an Integer Program (IP)
- Find decreasing sequence of upper bounds

$$\bar{z}_1 > \bar{z}_2 > \dots > \bar{z}_s \geq z$$

- Find increasing sequence of lower bounds

$$\underline{z}_1 < \underline{z}_2 < \dots < \underline{z}_t \leq z$$

- Algorithm stop when $\bar{z}_s - \underline{z}_t \leq \varepsilon$

RELAXATION AND BOUND

- Primal bounds
 - Every feasible solution $x^* \in X$ provides a lower bound of the maximization problem: $\underline{z} = cx^* \leq z$
 - Example: in TSP, every closed tour is a upper bound of the objective function (as TSP is a minimization problem)

RELAXATION AND BOUND

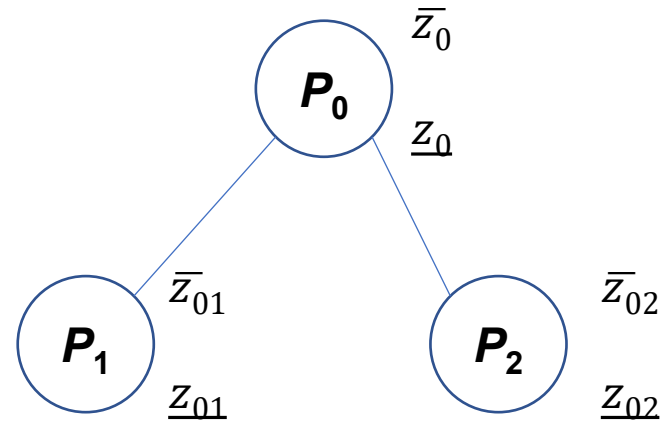
- Dual bounds
 - Finding upper bounds for a maximization problem (or lower bounds for a minimization problem) gives dual bounds of the objective
- **Definition** A problem $(RP) z^R = \max\{f(x): x \in T \subseteq R^n\}$ is a relaxation of $(IP) z = \max\{c^T x: x \in X \subseteq Z^n\}$ if:
 - $X \subseteq T$
 - $f(x) \geq c^T x, \forall x \in X$

RELAXATION AND BOUND

- Linear Relaxation
 - $Z^{LP} = \max\{c^T x : x \in P\}$ with $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ is a linear relaxation program of the (IP) $\max\{c^T x : x \in P \cap \mathbb{Z}^n\}$

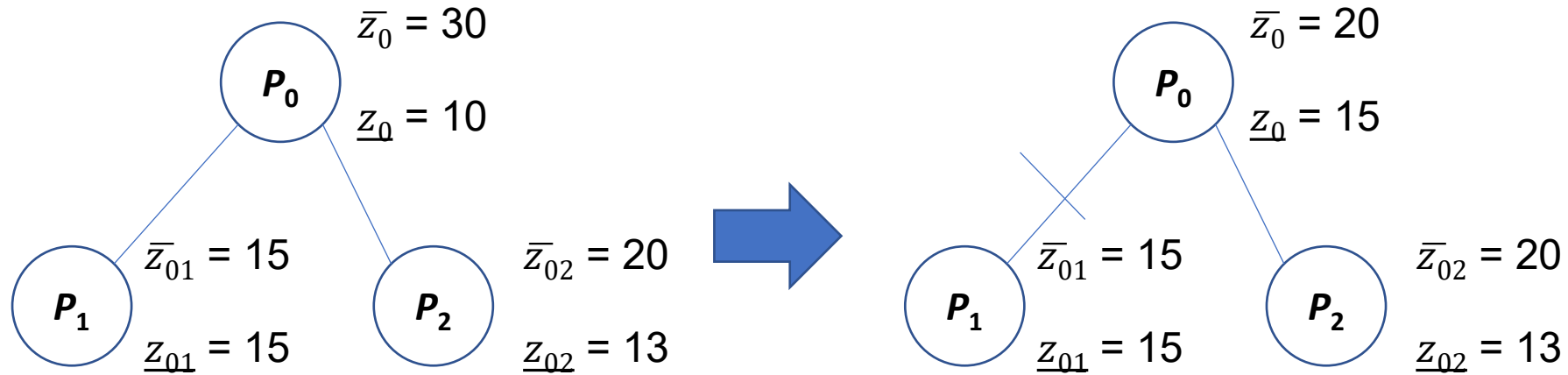
BRANCH AND BOUND

- Feasible region of P_0 is divided into feasible regions of P_1 and P_2 : $X(P_0) = X(P_1) \cup X(P_2)$



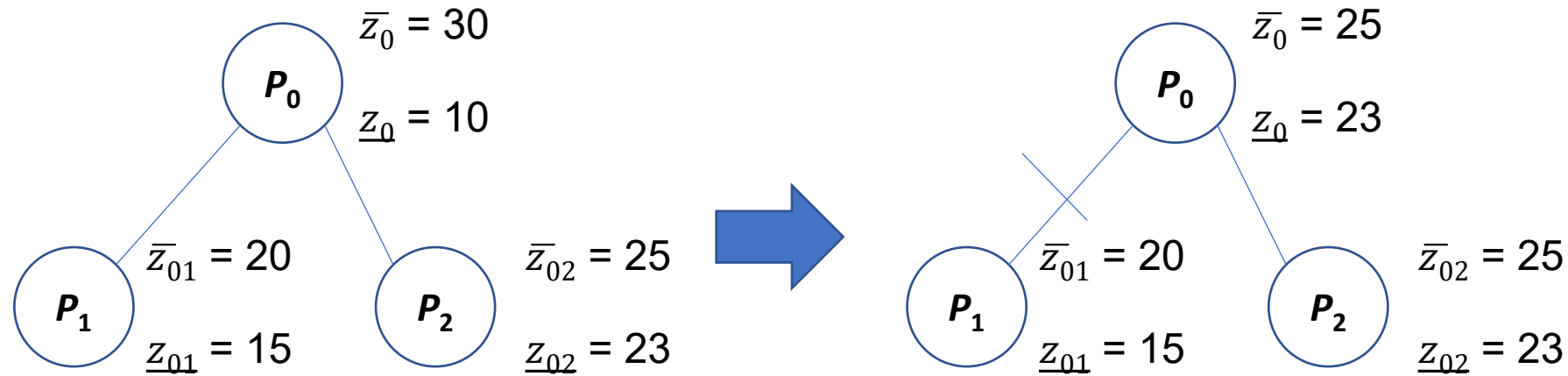
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BRANCH AND BOUND

- Branch and Bound schema

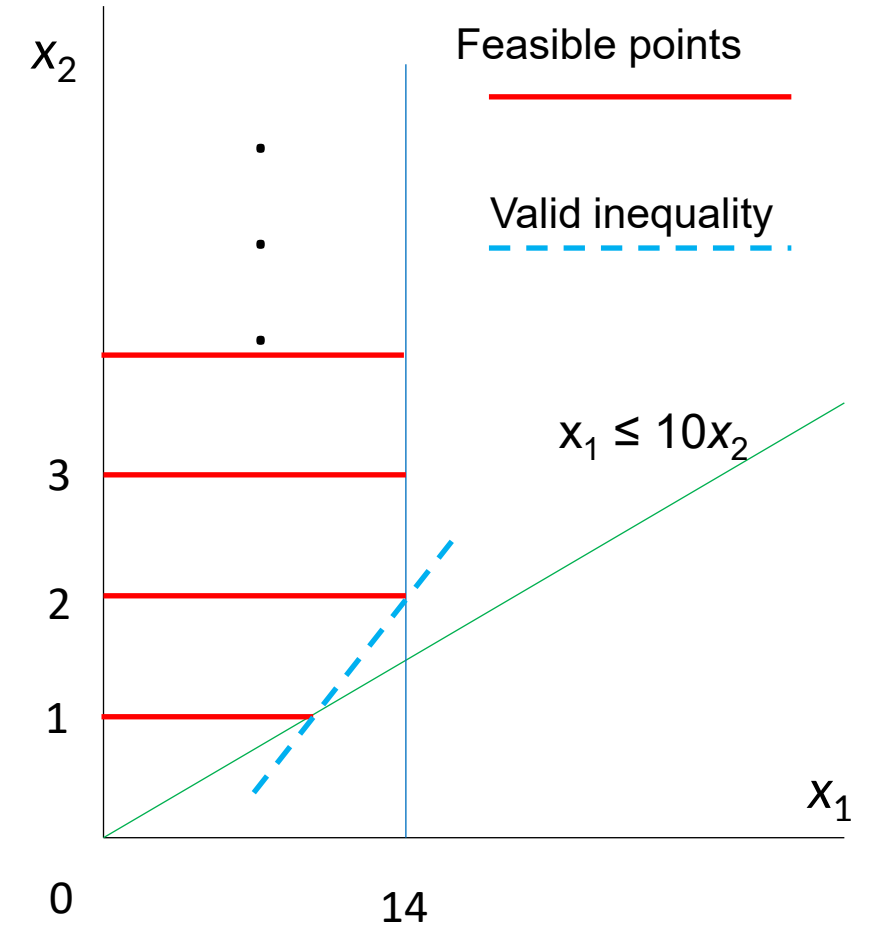
```
Initial problem  $S$  with formulation  $P$  on a list  $L$ ;  
Incumbent  $x^*$  is initialized with primal bound  $\underline{z} = -INF$ ;  
while  $L$  not empty do {  
    Select a problem  $S^i$  with formulation  $P^i$  from  $L$ ;  
    Solve LP relaxation over  $P^i$  got dual bound  $\bar{z}^i$  and solution  $x^i(LP)$ ;  
    if  $\bar{z}^i \leq \underline{z}$  then continue; // prune by dual bound;  
    if  $x^i(LP)$  integer then{  
         $\underline{z} = \bar{z}^i$  ;  
         $x^* = x^i(LP)$ ;  
    }else{  
        select a component  $x_i$  of  $x^i(LP)$  whose value  $\lambda_i$  is fractional;  
         $P_1^i = P^i \cup (x_i \leq \lfloor \lambda_i \rfloor)$ ,  $P_2^i = P^i \cup (x_i \geq \lceil \lambda_i \rceil)$ ;  
        add  $P_1^i$  and  $P_2^i$  to  $L$ ;  
    }  
}  
return  $x^*$ ;
```

CUTTING PLANE

- Given a (MIP) $\max\{c^T x: x \in X\}$
- Inequality $\pi x \leq \pi_0$ is called a valid inequality if $\pi x \leq \pi_0$ is true for all $x \in X$
- Finding valid inequalities allows us to narrow the search space, transform the (MIP) to a corresponding (LP) in which an optimal solution to (LP) is an optimal solution to the original (MIP)

CUTTING PLANE

- Example, consider a MIP with $X = \{(x_1, x_2): x_1 \leq 10x_2, 0 \leq x_1 \leq 14, x_2 \in \mathbb{Z}_+^1\}$
- Red lines represent X
- $x_1 \leq 6 + 4x_2$ is a valid inequality (dashed line)



CUTTING PLANE

- Example: **Integer Rounding**

- Consider feasible region $X = P \cap \mathbb{Z}^3$ where $P = \{x \in \mathbb{R}_+^3 : 5x_1 + 9x_2 + 13x_3 \geq 19\}$

- From $5x_1 + 9x_2 + 13x_3 \geq 19$ we have $x_1 + \frac{9}{5}x_2 + \frac{13}{5}x_3 \geq \frac{19}{5}$

$$\rightarrow x_1 + 2x_2 + 3x_3 \geq \frac{19}{5}$$

- As x_1, x_2, x_3 are integers, so we have

$$x_1 + 2x_2 + 3x_3 \geq \left\lceil \frac{19}{5} \right\rceil = 4 \text{ (this is a valid inequality for } X)$$

GOMORY CUT

- (IP) $\max \{cx: Ax = b, x \geq 0 \text{ and integer}\}$
- Solve corresponding linear programming relaxation (LP) $\max \{cx: Ax = b, x \geq 0\}$
- Suppose with an optimal basis, the (LP) is rewritten in the form

$$\begin{aligned}\overline{a_{00}} + \sum_{j \in J_N} \overline{a_{0j}} x_j &\rightarrow \max \\ x_{B_u} + \sum_{j \in J_N} \overline{a_{uj}} x_j &= \overline{a_{u0}}, u = 1, 2, \dots, m \\ x &\geq 0 \text{ and integer}\end{aligned}$$

with $\overline{a_{0j}} \leq 0$ (as these coefficients corresponds to a maximizer), and $\overline{a_{u0}} \geq 0$

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$$x_{B_u} + \sum_{j \in J_N} \overline{a_{uj}} x_j = \overline{a_{u0}}, u = 1, 2, \dots, m$$

$$x \geq 0 \text{ and integer}$$

with $\overline{a_{0j}} \leq 0$ (as these coefficients corresponds to a maximizer), and $\overline{a_{u0}} \geq 0$

- If the basic optimal solution x^* is not integer, then there exists some row u with $\overline{a_{u0}}$ is not integer

→ Create a Gomory cut $x_{B_u} + \sum_{j \in J_N} \lfloor \overline{a_{uj}} \rfloor x_j \leq \lfloor \overline{a_{u0}} \rfloor$ (1)

- **Example.** Consider the Integer Linear Program (ILP)

$$\begin{aligned} f(x_1, x_2, x_3, x_4, x_5) &= x_1 + x_2 \rightarrow \max \\ 2x_1 + x_2 + x_3 &= 8 \\ 3x_1 + 4x_2 + x_4 &= 24 \\ x_1 - x_2 + x_5 &= 2 \\ x_1, x_2, x_3, x_4, x_5 &\geq 0 \text{ and integer} \end{aligned}$$

- Solve the corresponding (LP)

$$\begin{aligned}f(x_1, x_2, x_3, x_4, x_5) &= x_1 + x_2 \rightarrow \max \\2x_1 + x_2 + x_3 &= 8 \\3x_1 + 4x_2 + x_4 &= 24 \\x_1 - x_2 + x_5 &= 2 \\x_1, x_2, x_3, x_4, x_5 &\geq 0\end{aligned}$$

→ Optimal solution (1.6, 4.8, 0, 0, 5.2) with $J_B = (1, 2, 5)$ and $J_N = (3, 4)$

Rewrite the original (ILP)

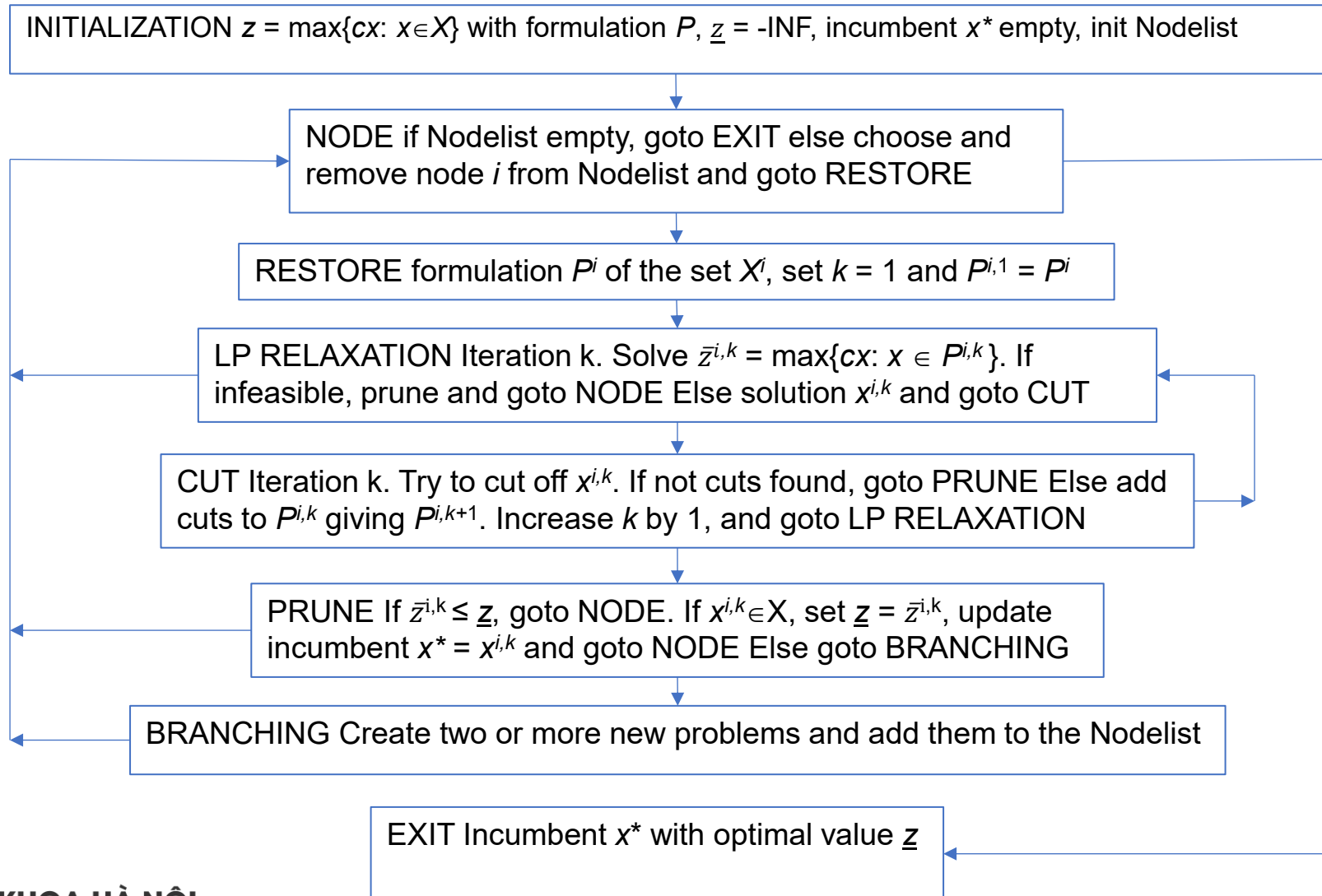
$$\begin{aligned}f(x_1, x_2, x_3, x_4, x_5) &= 6.4 - 0.2x_3 - 0.2x_4 \rightarrow \max \\x_1 + 0.8x_3 - 0.2x_4 &= 1.6 \\x_2 - 0.6x_3 + 0.4x_4 &= 4.8 \\x_5 - 1.4x_3 + 0.6x_4 &= 5.2 \\x_1, x_2, x_3, x_4, x_5 &\geq 0 \text{ and integer}\end{aligned}$$

GOMORY CUT

- Add gomory cut

- $x_1 + 0.8x_3 - 0.2x_4 = 1.6 \rightarrow$ add gomory cut: $x_1 + 0x_3 - x_4 \leq 1$
- $x_2 - 0.6x_3 + 0.4x_4 = 4.8 \rightarrow$ add gomory cut: $x_2 - x_3 + 0x_4 \leq 4$
- $x_5 - 1.4x_3 + 0.6x_4 = 5.2 \rightarrow$ add gomory cut: $x_5 - 2x_3 + 0x_4 \leq 5$

BRANCH AND CUT [Wolsey, 98]



A graphic on the left side of the slide. It features a dark blue background with a large, stylized circular shape composed of many small red dots. The dots are arranged in a way that creates a sense of depth and movement, with some dots appearing larger and more concentrated than others. The word "HUST" is written in white, bold, sans-serif capital letters across the center of this graphic.

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THANK YOU !